Reading Environment and History through Tree Rings

日本語 | Choose a language

An introductory course in dendrochronology: eight 90-minute lessons. No previous training in tree-ring measurement or statistics is required.

Lesson 1 requires no preparation. Its videos are for after-class review and independent study. The videos are older slide-reading prototypes and may not match the current materials.

Lessons | Supplementary resources | Offline use

Lessons

Each lesson's details page links to video chapters, exercise data and related resources. PDFs can be opened directly.

Lesson 1: Dating the past and reconstructing environments

Why dates matter and how different dating methods work.

Lesson 1 slides (PDF) · Lesson 1 handout (PDF) · Lesson 1 details and videos

Lesson 2: Tree growth, wood anatomy and tree-ring structure

Wood tissues, ring boundaries and the limits of counting.

Lesson 2 slides (PDF) · Lesson 2 handout (PDF) · Lesson 2 details and videos

Lesson 3: Principles of dendrochronology

Shared growth patterns and crossdating.

Lesson 3 slides (PDF) · Lesson 3 handout (PDF) · Lesson 3 details and videos

Lesson 4: Sampling, measurement and ring-width series

Documenting specimens, imaging, measuring and checking series.

Lesson 4 slides (PDF) · Lesson 4 handout (PDF) · Lesson 4 details and videos

Lesson 5: Dendroarchaeology

From ring formation to felling, construction and human history.

Lesson 5 slides (PDF) · Lesson 5 handout (PDF) · Lesson 5 details and videos

Lesson 6: Dendroecology

Suppression, release and disturbance in forest histories.

Lesson 6 slides (PDF) · Lesson 6 handout (PDF) · Lesson 6 details and videos

Lesson 7: Dendroclimatology

Climate response, calibration, verification and reconstruction limits.

Lesson 7 slides (PDF) · Lesson 7 handout (PDF) · Lesson 7 details and videos

Lesson 8: Research questions linking environment and history

Compare evidence and develop a research plan or literature review.

Lesson 8 slides (PDF) · Lesson 8 handout (PDF) · Lesson 8 details and videos

Supplementary resources

Use these resources as needed. The mathematics and statistics guide provides optional equations and background.

- Background disciplines (PDF)
- Disturbance and tree responses (PDF)
- Mathematics and statistics (PDF)
- Imaging wood for tree-ring measurement (PDF)
- Finding research papers and using DOIs (PDF)
- Observing wood anatomy in micrographs (PDF)
- Tree-ring glossary
- References and databases
- Use and credits

Offline use

Download the older course package (ZIP, including videos)

Extract the ZIP and open START_HERE.html in the en folder. The package is from 30 October 2026 and does not include later updates shown above.